AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) – 6

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Project Title: WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Results

4.1 Data Volume and Coverage

Table 1 summarises the data processed by the system as of this reporting period. The live WQMIS crawler has retrieved and scored 112 individual laboratory reports across Uttar Pradesh and Bihar to date. It should be noted that this represents an early, partial crawl rather than exhaustive state coverage: the WQMIS report-identifier space is large, and dense-range crawling has so far mapped only a fraction of it. Consequently, the live-sample results reported below should be read as a proof-of-concept sample rather than a statistically complete picture of contamination across either state. The supplementary historical contamination registry, by contrast, provides substantially broader coverage, contributing 100,690 quality-affected habitation events across 14,006 tracked villages for the 2009–2012 period, and forms the primary basis for the recurrence analysis in Section 4.3.

Metric	Value
Live WQMIS samples retrieved and scored	112 (Uttar Pradesh and Bihar)
Historical contamination registry events loaded (2009–2012)	100,690
Villages tracked in geographic hierarchy	14,006
Districts represented in choropleth mapping	121
Automated test cases passing	26 / 26
REST API endpoints exposed	9
Recurrence-query optimisation	37 s → 0.58 s (identical output)

4.2 Contamination Severity Findings

Applying the BIS 10500-based scoring engine (Section 3.3) to the retrieved live samples surfaced at least one Critical-band result during this period: a sample from Lucknow, Uttar Pradesh returned an E. coli detection, which under the band-escalation rule is classified as Critical irrespective of its composite numeric score, consistent with the rule's intent of not allowing an acute microbiological hazard to be diluted by unaffected parameters in the same report. Samples from Bhagalpur, Bihar showed the highest sample density observed so far (75 of the 112 retrieved samples), which is a function of where the crawler's mapped identifier ranges currently fall rather than a claim about Bhagalpur's relative contamination burden; this distinction is discussed further in Section 4.4.

4.3 Recurrence and Chronic Hotspot Findings

The recurrence detection module was run against the full historical registry to classify village-parameter combinations into the four verdict categories defined in Section 3.4. Among the results produced, the village of Khokha in Purnia district, Bihar, was flagged as a chronic-hotspot for iron contamination, with 166 recorded exceedance events across the historical registry period — the type of result the recurrence module is specifically designed to surface, since a single-reading severity score alone would not distinguish this village from one with a single isolated iron exceedance. A full quantitative breakdown of verdict-category counts (chronic-hotspot / recurring / watchlist / no-history) across all 14,006 villages is planned as an extension of this analysis in the final report.

4.4 Comparative Observations (Uttar Pradesh vs. Bihar)

The district-level choropleth view, covering 121 mapped districts across both states, allows visual comparison of sample density and contamination severity by state. At the current, partial stage of live-data crawling, direct state-to-state contamination comparison is limited by uneven crawler coverage (for example, the concentration of samples in Bhagalpur relative to other districts), and any comparative claims at this stage describe where data has been collected so far rather than underlying contamination differences between the two states. This is a known limitation of the current results, distinct from the historical registry analysis in Section 4.3, which draws on materially more complete data and therefore supports comparison with greater confidence.

4.5 System Performance and Validation Results

All twenty-six automated test cases pass, covering scoring correctness, API contract behaviour, WQMIS parser edge cases, and recurrence logic including a PostgreSQL integration suite. The nine REST endpoints and the three-view dashboard (priority queue, state comparison, district map) were verified to function correctly against both the live and offline-snapshot data paths. The recurrence-ranking performance issue identified during development (Section 3.7) was resolved, reducing execution time from approximately 37 seconds to 0.58 seconds across the full 14,006-village set, with output verified identical to the original implementation via regression tests.